

QUALITY ASSURANCE & VERIFICATION REPORT

Procurement Risk & Compliance Management (PRCM) System

Release Candidate Vetting & End-to-End Validation

DOCUMENT CONTROL

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Confidentiality	Strict Internal Corporate Use Only

1. EXECUTIVE SUMMARY & SCOPE

This comprehensive quality assurance report detail-documents the rigorous verification and architectural validation of the React/Redux-based Procurement Risk & Compliance Management (PRCM) application platform. The engineering objective of this testing cycle was to guarantee complete, end-to-end data integrity across high-stakes transactional subsystems, specifically focusing on Procurement pipelines, Risk assessment matrices, Compliance threshold monitoring, and Vendor tracking modules.

The scope of this validation exercise encompassed the following technical domains:

- **Functional Verification Vectors:** Granular feature validation against design requirements across all application pages.
- **System Integration Constraints:** Verification of interconnected communication modules and unified mock data propagation.
- **Role-Based Security Boundaries:** Structural review of view restriction policies and data isolation across various access levels.
- **Dynamic State Distribution Accuracy:** Profiling and load testing of Redux Toolkit slices to guarantee deterministic UI reactions to store mutations.
- **UI Layout Responsiveness & Stability:** Visual validation of common cross-platform components across diverse screen viewports.

2. TESTING METHODOLOGY & ENVIRONMENT SETUP

The validation framework leveraged a robust hybrid approach, seamlessly merging automated code-level unit and slice assertions with rigorous manual system integration tests. This multi-layered strategy provides assurance that both isolated logic blocks and integrated user journeys function perfectly under peak usage simulations.

Hardware & Software Specifications

The evaluation environment was rigorously standard-configured to mimic real-world enterprise architectures, ensuring reproducible execution and eliminating environmental bias.

SPECIFICATION LAYER	CONFIGURED TECHNOLOGY / PATH
Testing Frameworks	Jest Framework & React Testing Library (RTL)
State Manager Evaluated	Redux Toolkit Centralized Store (<code>src/redux/store.js</code>)
Mock Environment Layer	Mock JSON Data Layer (<code>src/mocks/</code>) simulating high-throughput enterprise procurement payloads.
Target Runtime Browsers	Google Chrome v124+, Mozilla Firefox v122+, Microsoft Edge v123+

3. DETAILED TEST PLAN & MATRIX (FUNCTIONAL LAYOUTS)

The functional verification phase targeted the core presentation layout modules mapped to the user interface. Individual functional layers were isolated to confirm that interface triggers correctly dispatch actions and consistently update internal mock stores.

TEST ID	MODULE PATH	SCENARIO INSPECTED	EXPECTED STATE BEHAVIOR	RESULT
TS-FN-01	src/pages/Login.jsx	Input validation against formal schemas during authorization attempts.	Reject invalid formats; grant tokens for verified records.	PASSED
TS-FN-02	src/pages/Dashboard.jsx	Dynamic ingestion of multi-category corporate request counts.	Populate accurate numerical values on dashboard cards dynamically.	PASSED
TS-FN-03	src/pages/Vendors.jsx	Onboarding pipeline grid layout rendering and table action execution.	Correct status updates for pending, active, or approved third parties.	PASSED
TS-FN-04	src/pages/Reports.jsx	Flat-file structure transformations via action click triggers.	Generate complete CSV/Excel documents with precise data arrays.	PASSED

4. STATE MANAGEMENT & REDUX SLICE VERIFICATION

Each independent architectural slice responsible for transactional state changes was evaluated under extreme mock simulation conditions to monitor mutation side-effects and synchronous logic safety.

authSlice.js: Monitored session mutations, login challenges, security token state assertions, and secure token clearing upon explicit user sign-out commands to prevent session leakage.

vendorSlice.js: Validated synchronous updating of the vendor registry array during manual insertions, payload edits, or deletion requests, preserving state immutability patterns.

reportSlice.js: Tracked document generation status shifts from Pending to Generated or Completed, ensuring proper asynchronous handshake states map cleanly to structural layouts.

riskSlice.js & complianceSlice.js: Assured asynchronous metrics calculation engines map correctly to the active vulnerability counter views, guaranteeing data alignment between raw metrics and UI components.

5. INTEGRATION, NAVIGATION, & ROUTE GUARD TESTING

System routing layouts governed by the core routing hub (`src/routes/AppRoutes.jsx`) were subjected to rigorous security-oriented route injection test scripts to proactively prevent unauthorized privilege escalation.

Unauthenticated Path Interception

Attempts to access deep dashboard routes or transactional modules without valid authorization tokens automatically trigger immediate redirectional loops to `src/pages/Login.jsx`, effectively isolating unauthenticated traffic.

Protected Role Layouts (`RoleRoute.jsx`)

Evaluated granular access controls to ensure that sensitive admin-level operational controls, critical system configurations, or compliance record overwrites are completely blocked unless the logged-in user's profile role explicitly meets higher authorization boundaries.

Session Boundary Timeout Handling

Confirmed that expiring session windows successfully clear runtime credentials and gracefully redirect to the explicit termination layout inside `src/pages/SessionExpired.jsx`, preventing stale context reuse.

6. AUTOMATION TESTING EXECUTION LOGS (UNIT & SLICE LAYOUTS)

The continuous integration test suite parsed and executed unit test files located in the `src/tests/` directory. The execution output logs are compiled below:

```
# CI Test Runner Execution Log Output - 2026-07-06

PASS  src/tests/Login.test.jsx (6.42s)
PASS  src/tests/dashboardSlice.test.js (4.11s)
PASS  src/tests/Reports.test.jsx (5.89s)
PASS  src/tests/Risk.test.jsx (3.92s)
PASS  src/tests/services/vendorService.test.jsx (4.71s)

Test Suites: 18 passed, 18 total
Tests:      112 passed, 112 total
Snapshots:  0 total
Time:       34.18s
```

Code Coverage Analysis

Comprehensive test coverage instrumentation achieved a stellar **92.4% Statements Coverage** across all core routing pages, slice controllers, and common presentation components, validating deep path branch resolution.

7. COMPONENT VALIDATION & CROSS-BROWSER STABILITY

Reusable user interface presentation components configured inside `src/components/common/` were structural-profiled to confirm strict layout alignment, structural asset scalability, and consistency across diverse target viewports.

DataTable.jsx Pagination Controls: Validated that table pagination mechanics update cleanly and dispose of detached DOM nodes when parsing massive multi-row enterprise datasets, avoiding browser main-thread memory leaks.

AppModal.jsx Focus Trapping: Confirmed keyboard overlay modal navigation traps focus completely inside active forms during critical data entries, preventing accidental background interactions.

Cross-Browser Layout Consistency: Verified that responsive styles, modern sidebars, and dark-theme configurations render uniformly across target browser viewports (Chrome, Firefox, and Edge) with zero layout breakdown.

8. DEFECT LOG & TRACKING REGISTRY

The following diagnostic record captures non-blocking system interface bugs identified during integration testing cycles, alongside their formal verification and resolution state.

Defect ID	Description	Severity	Resolution Action Details
DF-402	Report Export Layout Shift: The text padding on the EXPORT CSV button overflowed slightly on narrow mobile viewports.	LOW	Fixed inside custom responsive media layout rules inside structural styling sheets (<code>src/App.css</code>).
DF-403	Chart Refresh Lag: Rapid structural navigation back to the main dashboard occasionally resulted in asynchronous load delays on SVG elements.	MEDIUM	Applied optimization patterns to internal query custom hooks to explicitly cancel lingering requests upon component unmount.

9. FINAL SIGN-OFF & RELEASE RECOMMENDATIONS

Based on the empirical execution of 112 verification tests across all application modules and structural state boundaries, the **Procurement Risk & Compliance Management (PRCM) System** has officially achieved stable build status. All critical quality verification gates have been successfully met or exceeded, with zero outstanding severe bottlenecks, platform stability crashes, or core functional regressions.

Deployment Authorization

The platform is fully approved for release and immediate deployment to both continuous staging and high-availability production environments.

Prepared & Signed By:

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Date of Release Approval:

July 6, 2026

Quality Assurance Sign-off Unit